

# Yanghao Chen

Tongji University, Shanghai, China | [2350083@tongji.edu.cn](mailto:2350083@tongji.edu.cn) | [Homepage](#)

## Education

### Tongji University

Bachelor of Engineering in Engineering Mechanics

GPA: 87/100

Shanghai, China

Sep. 2023 – Present; expected Jun. 2027

### Johns Hopkins University

Visiting Undergraduate Research Intern, Department of Civil and Systems Engineering

Baltimore, MD, USA

Jan. 2026 – Oct. 2026

**Research Interests:** Solid Mechanics, Scientific Machine Learning, Neural Operators.

## Research Projects

### GPU-Accelerated Hybrid FEM–Neural Operator Solver

Jan. 2026 – Present

Lead Undergraduate Researcher; Advisor: [Prof. Somdatta Goswami](#)

Johns Hopkins University

- Developed a hybrid FEM–neural operator solver based on non-overlapping domain decomposition, using FEM and neural operators in separate subdomains.
- Generated data using GPU-accelerated JAX-FEM simulations with prescribed interface displacements; trained Transolver in JAX to learn mappings from interface displacement fields to subdomain stress fields.
- Computed interface reaction forces from stresses predicted in the neural-operator subdomain; passed them to FEM and fed the updated interface displacements back to the neural operator at each coupling iteration.

### PINNs for Shock Capturing in the Burgers Equation

Nov. 2025 – Mar. 2026

Undergraduate Researcher; Advisor: [Prof. Xianyang \(Tom\) Chen](#)

Tongji University

- Trained a PINN in PyTorch to predict shock propagation in the one-dimensional inviscid Burgers equation, using PDE-residual, initial-condition, and boundary-condition losses.
- Added constant artificial viscosity to the PDE residual, using automatic differentiation to compute the required derivatives.
- Compared predicted shock profiles with the analytical inviscid solution, examining transition width and local deviations near the discontinuity.

### ConvLSTM Modeling of Chemo-Mechanical Fields in Battery Materials

Jun. 2025 – Oct. 2025

Undergraduate Researcher; Advisor: [Prof. Ying Zhao](#)

Tongji University

- Automated polycrystalline NMC microstructure generation and MATLAB–COMSOL simulations to produce time-aligned images of lithium concentration and von Mises stress.
- Trained separate conditional ConvLSTM models in PyTorch to predict future sequences of concentration and stress fields from past field images, grain-orientation maps, and C-rate inputs.
- Used MSE and SSIM losses with scheduled sampling and patch-based training for multi-step autoregressive prediction.

### Domain-Specific LLM Fine-Tuning for Mechanics of Materials

Sep. 2024 – Apr. 2025

Undergraduate Researcher; Advisor: [Prof. Ying Zhao](#)

Tongji University

- Curated instruction–response datasets from textbooks and technical literature on stress analysis, constitutive laws, and failure theories.
- Fine-tuned Qwen2.5-7B with LoRA on Google Colab for mechanics of materials Q&A and concept explanations.
- Released two instruction datasets and two Qwen2.5-based model checkpoints on [Hugging Face](#).

## Technical Skills

### Scientific Machine Learning

PyTorch, JAX, Neural Operators (Transolver, DeepONet), PINNs, ConvLSTM

### Computational Mechanics

Finite Element Methods, FEniCSx, Gmsh, COMSOL Multiphysics, ABAQUS

### Scientific Computing & Tools

Python, MATLAB, Linux, Git/GitHub, LiveLink for MATLAB

### LLM Fine-Tuning

Qwen2.5, LoRA/PEFT, Instruction Dataset Curation

### Languages

Chinese (native), Cantonese, English